

CSMFAB Materials Study — Part II

Advanced Composite Fibers, Resins, and Structural Non-Metallic Materials

Carrington Storm Motors | Safe Pod Engineering Company | June 2026

Study Scope: Fibers (basalt, UHMWPE, carbon), matrix resins (Elium®, epoxy, PEEK), aerogel systems, and piezoelectric polymers used across the CSMFAB portfolio.

Used In: Nearly ALL CSMFAB products (28 of 43 use BFRP/Elium® as primary structure)

Material 9 — Basalt Fiber

Source: Natural volcanic basalt rock | **Class:** Inorganic mineral fiber | **Key Property:** Non-conductive, non-magnetic

9.1 Physical Properties

- Tensile strength: 3,000–4,840 MPa (manufacturing-grade to premium)
- Elastic modulus: 89–95 GPa
- Density: 2.65–2.80 g/cm³
- Dielectric constant: 6.4–7.2 at 1 MHz — excellent insulator
- Operating temperature: -200°C to +700°C continuous (vs. glass fiber: max 350°C)
- **Electrical resistivity: >10¹² Ω·m — zero GIC coupling**
- Color: dark brown/black (raw fiber); can be coated

9.2 How Basalt Fiber is Made

Raw material: Any volcanic basaltic rock (SiO₂ 45–55%, Al₂O₃ 12–17%, CaO 8–11%, MgO 6–10%, FeOx 9–15%) **Abundance:** Basalt is the **most common rock on Earth's surface** — essentially free

Manufacturing process: 1. Mine basalt rock (quarry cost \$2–\$5/ton) 2. Crush to aggregate (\$1–\$2/ton processing) 3. Melt in tank furnace at 1350–1450°C (same furnace as glass fiber) 4. Draw through platinum-rhodium bushing tips (200–400 tips per bushing) 5. Apply sizing (coupling agent, typically silane-based) 6. Wind on bobbins

Critical difference from glass fiber: No additives (no boron, no rare earths) — just rock. **This makes basalt fiber potentially the cheapest performance fiber with domestic rock supply.**

9.3 Commercial Acquisition

Supplier	Country	Price/kg (2026)	Form
Mafic USA (Shannonbridge, Ireland; US subsidiary)	USA/Ireland	\$4.50–\$7.00	Roving, chopped, woven
Kamenny Vek	Russia (currently sanctions-limited)	\$3.00–\$5.00	Historical reference
Basaltex (Belgium)	Belgium	\$7.00–\$12.00	Specialty woven
Shanxi Basalt Fiber	China	\$2.50–\$4.00	Roving
Sudaglass (USA)	USA	\$8.00–\$14.00	Specialty applications

Aerospace grade (high tensile, consistent OD): \$15–\$35/kg

9.4 In-House Production of Basalt Fiber — HIGHLY FEASIBLE

The single most impactful in-house production opportunity in the CSMFAB portfolio.

Equipment: - Basalt rock crusher and grinder: \$15,000–\$40,000 - Electric resistance tank furnace (1450°C, 2-ton): \$85,000–\$180,000 - Platinum-rhodium bushings (key component): \$40,000–\$80,000 per bushing - Sizing applicator and winding machines: \$25,000–\$50,000 - **Total capital investment: \$165,000–\$350,000**

Raw material costs: - Local basalt rock (quarried): \$2–\$5/ton (essentially free) - Na₂CO₃ flux (optional, improves meltability): \$0.30/kg - Silane coupling agent: \$3.50/kg (100g per 100g fiber)

In-house cost estimate: - Energy: 4 kWh/kg fiber × \$0.08/kWh = \$0.32/kg - Labor: \$0.80/kg - Raw materials: \$0.15/kg - Equipment amortization (\$250K over 10 years, 500 ton/yr): \$0.50/kg - **Total in-house: \$1.80–\$2.50/kg vs. \$4.50–\$14.00 commercial - Savings: 60–85%**

Feasibility Rating: 5/5 — same process as glass fiber (proven industrial tech); domestic rock = zero raw material cost. This is the CSM portfolio's crown-jewel in-house production project.

U.S. domestic basalt sources with verified fiber-quality composition: - Columbia River Basalt (Oregon/Washington): excellent Fe, Ti balance - Snake River Plain (Idaho): high alumina grades preferred - Hawaii (Mauna Loa): ultra-clean oceanic basalt

9.5 Basalt Fiber for Specific CSMFAB Applications

Application	Fiber form	Target spec
BFRP structural panels	Unidirectional roving	4,000+ MPa, 2.70 g/cm ³
BFRP rebar	Rovings (pull-wound)	3,500 MPa, consistent OD ±0.1mm
Aerogel reinforcement	Chopped fiber (6mm)	3,200 MPa min
Ceramic-fiber hybrid	Chopped or milled (<200 µm)	Purity, no metallic inclusions
Textile for covers/upholstery	Woven 2/2 twill, 200g/m ²	Comfort sizing required

Material 10 — Elium® Thermoplastic Acrylic Resin (Arkema)

Chemical class: Liquid thermoplastic acrylic resin (methyl methacrylate base) | **Key property:** Recyclable at end of life

10.1 Why Elium® Is Central to the Phoenix Protocol

Standard fiber-reinforced composites use thermoset epoxy or polyester — once cured, irreversible. Elium® is a **thermoplastic resin that processes identically to thermoset** but can be: - Dissolved by gentle thermal treatment at 350°C □ monomer recovered (100% yield) - Fiber recovered simultaneously (95% basalt fiber yield) - Re-polymerized and re-used — **true circular composite manufacturing**

10.2 Chemical Composition and Properties

- Base monomer: methyl methacrylate (MMA) — same as Plexiglas
- Additives: initiator, chain transfer agent, rheology modifier
- Liquid viscosity: 100–300 mPa·s (compatible with infusion, VARTM, RTM)
- Cured Tg: 80–110°C (depending on cure profile)
- Tensile modulus (neat resin): 3.2 GPa
- Elongation at break: 6–12%
- Cure time: 30–120 minutes at room temperature (initiated by peroxide or UV)

Composite mechanical properties (with 60% basalt FVF): - Tensile strength: 1,000–1,100 MPa (fiber direction) - Modulus: 55–65 GPa

10.3 Raw Material: Methyl Methacrylate (MMA)

MMA commercial supply: - Rohm (Evonik): Dominant German producer - Mitsubishi Chemical: Japan - Trinseo: USA - **MMA price 2026: \$1.80–\$2.80/kg (bulk liquid)**

MMA synthesis: 1. **ACH (Acetone Cyanohydrin) process** (conventional): Acetone + HCN → methacrylamide → MMA 2. **C2 (Lucite Alpha) process** (modern, lower cost): Ethylene + CO + methanol → methyl propanoate → MMA 3. **Bio-based MMA:** From sugarcane-derived lactic acid or bio-propylene (2025 pilot — Arkema, Mitsubishi)

Can CSM produce MMA in-house? Not economically — MMA plant requires \$50M+ capital. **Action:** Establish long-term supply agreement with Arkema/Trinseo at volume pricing.

10.4 Commercial Elium® Pricing

Grade	Price/kg (2026)	Application
Elium® 150 (infusion)	\$8.50–\$12.00	VARTM, RTM
Elium® 188 (prepreg)	\$14.00–\$18.00	Prepreg layup
Elium® 175 (pultrusion)	\$9.00–\$13.00	Rebar, profiles
MMA monomer (bulk)	\$1.80–\$2.80	In-house formulation

In-house Elium® formulation from MMA: MMA + peroxide initiator (0.5–2%) + chain transfer agent → Elium-equivalent Equipment: mixing vessel + cold storage; minimal. Cost: **\$4–\$6/kg** (vs. \$8.50–\$18 commercial) **Feasibility Rating:** 4/5 — well-documented MMA polymerization; initiator sourced easily.

10.5 VARTM Processing of Basalt/Elium®

Vacuum-Assisted Resin Transfer Molding (VARTM): - Dry basalt preform placed in mold - Vacuum applied; Elium® drawn in under atmospheric pressure - Single-sided tooling only (open mold sufficient) - Cure at room temperature: no heated press required - **Equipment: \$15,000–\$35,000** (molds, vacuum pump, infusion lines) - Production rate: 2–8 parts/day per mold (depending on part complexity)

Products using BFRP/Elium®: Virtually all structural CSMFAB products (001, 003, 004, 005, 006, 007, 009, 011–014, 019, 024, 028–040, 050, CSMIC 105, CSMMECH 001/003)

Material 11 — UHMWPE (Ultra-High Molecular Weight Polyethylene) — Dielectric Rope

Chemical class: Polyethylene, Mw >3.5 million g/mol | **Brand names:** Dyneema® (DSM), Spectra® (Honeywell)

11.1 Physical Properties

- Tensile strength: 3,500–4,250 MPa (fiber, SK99 grade)
- Density: 0.97 g/cm³ (floats on water)
- **Electrical resistivity: >10¹⁴ Ω·m — perfect GIC insulator**
- UV resistance: moderate (requires UV stabilizer)
- Cut resistance: 15× steel at equal weight

11.2 Raw Material Source

Precursor: Ethylene gas (C₂H₄) — from petroleum refining or bioethanol dehydration **Polymerization:** Ziegler-Natta or metallocene catalysis □ UHMWPE **Gel spinning (fiber production):** UHMWPE dissolved in solvent □ extruded □ solvent extracted □ oriented

Bio-based UHMWPE (emerging, 2025): - Sugarcane bioethanol □ ethylene □ UHMWPE - Braskem (Brazil) “Green PE” — from 100% sugarcane; carbon-negative - Price premium: +\$1.50–\$3.00/kg over petroleum UHMWPE

11.3 Commercial Pricing

Grade	Price/kg (2026)	Notes
UHMWPE granules (bulk)	\$4.50–\$7.50	Sheet/rod production
Dyneema® SK75 fiber	\$18–\$28	Rope, ballistic
Dyneema® SK99 fiber	\$35–\$55	Highest tenacity
Spectra® 1000	\$25–\$40	Honeywell equivalent
Braskem Green PE	\$7.00–\$10.00	Bio-based granules

11.4 In-House Processing: Rope Braiding

Gel-spinning fiber from UHMWPE granules requires \$1–5M specialized equipment — not feasible in-house. **Action: Purchase Dyneema® SK75/SK99 fiber at volume pricing and braid in-house.**

Rope braiding equipment: \$12,000–\$35,000 (16-carrier horizontal braider) In-house braiding adds: \$1.20–\$2.50/m of rope (labor + amortization) **Total rope cost: \$4–\$8/m (12mm diameter) vs. \$12–\$20/m finished product**

Products: 017 (rope kit), 019 (forklift rigging), 050 (drone harness)

Material 12 — Aerogel Systems

Types used in CSMFAB: 1. Hydrophobic silica aerogel (baseline, historical) 2. **Polyimide-silica hybrid aerogel (V2.0 specification** — $\lambda=0.010$ W/m·K, 650°C service) 3. Cellulose nanocrystal (CNC) aerogel (bio-based, non-toxic, for Embark pet-safe products)

12.1 Silica Aerogel — The Original

Raw materials: - Tetraethyl orthosilicate (TEOS) or sodium silicate (water glass): \$2–\$4/kg - Methanol or ethanol: \$0.60–\$1.20/kg - MTMS (methyltrimethoxysilane): hydrophobizing agent \$8–\$15/kg - HF or NH₄F catalyst (trace)

Sol-gel synthesis: TEOS + ethanol + water + acid/base □ silica alcogel □ atmospheric drying (with MTMS) □ aerogel

Commercial aerogel costs: - Cabot Enova® particle aerogel: \$5–\$8/kg (bulk) - Aspen Aerogels Pyrogel® XT (blanket form): \$120–\$180/m² (10mm thickness) - Aspen Aerogels Pyrogel® XT-E (polyimide-silica): \$200–\$280/m² - Nano Hi-Tech (China): \$35–\$65/m² (lower performance)

12.2 Polyimide-Silica Hybrid Aerogel — V2.0 Critical Material

Synthesis route: 1. Silica sol (TEOS-derived) + polyimide precursors (PMDA-ODA or BPDA-ODA) mixed 2. Gelation □ polyimide crosslinks reinforce silica network 3. Supercritical CO₂ drying □ hybrid aerogel 4. Or: ambient-pressure drying (with progressive solvent exchange) — 70% energy savings

Performance gains over pure silica aerogel: - λ : 0.010 vs. 0.015 W/m·K - Max service: 650°C vs. 600°C - Flexural strength: 3-4× (less brittle in handling)

Commercial hybrid aerogel: \$200–\$280/m² (Aspen Aerogels, 2026)

In-house synthesis feasibility: - TEOS: \$4/kg (bulk chemical) | Polyimide precursor (PMDA): \$25–\$40/kg - Synthesis equipment: gel casting setup, supercritical CO₂ dryer - Supercritical dryer (30L capacity): \$35,000–\$80,000 — **major capital cost** - Or: ambient-pressure dryer (modified) with TMCS solvent exchange: \$8,000–\$15,000

In-house cost estimate: \$55–\$85/m² (30mm blanket) vs. \$200–\$280/m² commercial **Savings: 65-75%**
Feasibility Rating: 3/5 — supercritical dryer is expensive but shared with entire aerogel product line. Ambient-pressure route is more accessible.

12.3 CNC (Cellulose Nanocrystal) Aerogel

Raw material: Wood pulp, cotton, agricultural waste (any cellulose source) - Acid hydrolysis with H₂SO₄ □ CNC suspension □ freeze-dry or CO₂ dry - Bio-based, non-toxic, carbon-negative - Performance: λ =0.018 W/m·K (slightly less than silica hybrid); excellent for comfort/pet-safe applications

Commercial: \$120–\$200/m² (limited commercial production; Oak Ridge ORNL technology) **In-house:** Agricultural waste + H₂SO₄ treatment + freeze dryer = **\$25–\$45/m²** Freeze dryer capital: \$15,000–\$40,000
Feasibility Rating: 4/5 — simple processing; agricultural waste is free or nearly free.

Products using aerogel: 003, 004, 006, 007, 008, 011, 012, 014, 024, 028, 030, 036, 037

Material 13 — PVDF-TrFE Piezoelectric Polymer Film

Full name: Poly(vinylidene fluoride-trifluoroethylene) | **Class:** Ferroelectric semicrystalline polymer

13.1 Physical/Electrical Properties

- d₃₃: -38 to -50 pC/N (longitudinal piezoelectric coefficient)
- d₃₁: +21 to +28 pC/N
- Curie temperature: 55–65°C (PVDF-TrFE-CFE grade for electrocaloric: 80°C)
- Dielectric constant: 8–12 at 1 kHz
- Operating range: -40°C to +100°C

13.2 Synthesis and Raw Materials

Precursors: - VDF monomer (vinylidene fluoride): from vinylidene chloride via fluorination or from 1,1,1-trifluoro-2-chloroethane. \$3–\$5/kg (industrial) - TrFE monomer (trifluoroethylene): from acetylene + HF; \$8–\$15/kg - Optionally: CTFE or HFP for terpolymer grades

Synthesis: Bulk or suspension polymerization in pressure reactor □ PVDF-TrFE copolymer □ solution cast into film □ stretch/pole to align ferroelectric domains

Commercial suppliers: - Piezotech/Arkema (France): \$180–\$280/kg film resin; \$250–\$450/m² film - Measurement Specialties (TE Connectivity): \$200–\$350/m² film - Solvay Solef: PVDF base resin \$8–\$12/kg (not PVDF-TrFE)

13.3 In-House Production

Fluorinated monomers require specialized pressure equipment and HF handling — **high safety barrier.**
Recommendation: Purchase PVDF-TrFE resin from Arkema/Piezotech and cast films in-house.

Film casting equipment: \$25,000–\$50,000 (film casting line, stretching frame, poling electrodes) In-house cost: **\$80–\$120/m²** (from commercial resin) vs. \$250–\$450/m² fully commercial film **Feasibility Rating: 3/5** for film casting in-house (from purchased resin).

Products: 004 (electrocaloric cooling), 024 (footwear harvesting), 026 (pedal harvesting), 028, 038, MAG (Bio-Sync)

Material 14 — h-BN (Hexagonal Boron Nitride)

Formula: BN hexagonal | **Class:** 2D van der Waals ceramic / thermal conductor + electrical insulator

14.1 Properties

- Thermal conductivity: 250–400 W/m·K (in-plane basal)
- Electrical resistivity: $>10^{14} \Omega\cdot\text{m}$
- Dielectric strength: 6 kV/mm
- Max service: 900°C in air; 2000°C in inert atmosphere
- Lubricity: similar to graphite (hexagonal structure)

14.2 Synthesis

Routes: 1. $\text{B}_2\text{O}_3 + \text{NH}_3$ at 800–1000°C $\rightarrow$ BN (most common commercial) 2. $\text{BCl}_3 + \text{NH}_3 \rightarrow$ BN (gas-phase CVD) 3. Boric acid + urea $\rightarrow$ at 800°C $\rightarrow$ h-BN (simple, low-cost)

Raw materials: - Boric acid (H_3BO_3): Borax ($\text{Na}_2\text{B}_4\text{O}_7$) + $\text{H}_2\text{SO}_4 \rightarrow \text{H}_3\text{BO}_3$; \$0.80–\$1.50/kg - Urea: nitrogen fertilizer \$0.35–\$0.60/kg - Or: NH_3 gas \$0.25–\$0.50/kg

14.3 Commercial Pricing

Grade	Price/kg (2026)
Powder (1-5 μm)	\$28–\$45/kg
Nanoplatelets (<200 nm)	\$80–\$150/kg
Film/coating (spray)	\$15–\$25/m ²

14.4 In-House Synthesis

Boric acid + urea method: Equipment: tube furnace (1000°C, NH_3 atmosphere available or urea provides N-source) In-house cost: **\$8–\$15/kg** from commodity boric acid + urea **Feasibility Rating: 4/5** — simple synthesis; uses inexpensive commodity precursors.

Used in: 013 (wiring insulation dielectric filler), CSMMECH 001 (ceramic-coated wiring)

Summary Table — Part II Composite and Fiber Materials

Material	Commercial (\$/kg or /m ²)	In-House Estimate	Savings	Rating
Basalt fiber	\$4.50–\$35/kg	\$1.80–\$2.50/kg	60–85%	5/5
Elium® from MMA formulation	\$8.50–\$18/kg	\$4–\$6/kg	40–65%	4/5

Material	Commercial (\$/kg or /m ²)	In-House Estimate	Savings	Rating
UHMWPE fiber (rope braiding)	\$18–\$55/kg	buy fiber + braid	40–55% vs. rope	3/5
Silica aerogel blanket	\$120–\$280/m ²	\$55–\$85/m ²	55–75%	3/5
CNC cellulose aerogel	\$120–\$200/m ²	\$25–\$45/m ²	75–80%	4/5
PVDF-TrFE film (cast in-house)	\$250–\$450/m ²	\$80–\$120/m ²	65–75%	3/5
h-BN powder	\$28–\$150/kg	\$8–\$15/kg	60–75%	4/5

Highest-priority in-house production: 1. **Basalt fiber** — highest volume use, 60-85% savings, domestic rock feedstock 2. **Aerogel blankets** — used in 14 products; shared supercritical/ambient-pressure dryer 3. **Basalt/Elium® BFRP panels** — VARTM processing using in-house fiber+resin

End of CSMFAB Materials Study Part II | Composite Fibers and Resins Carrington Storm Motors | June 2026